

SUMMARY

AI/ML Engineer building production ML and LLM systems with Python, PyTorch, FastAPI, Docker, and CI/CD. Experienced in taking projects from data preparation and model development through evaluation and deployment, with a focus on privacy preserving ML, synthetic data, and RAG applications.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, Java, JavaScript, Bash
ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV
GenAI & LLMs: RAG, Vector DBs, Prompt Engineering
Databases: MySQL, PostgreSQL, SQLite, MongoDB
DevOps & Tools: Docker, FastAPI, REST APIs, Git, CI/CD

PROFESSIONAL EXPERIENCE

AI Engineer

MyVellum

Dec. 2025 – Present

Remote

- Built and containerized a FastAPI RAG service with Groq-hosted LLMs, vector-backed retrieval, and modular prompt orchestration
- Implemented CI/CD evaluation gates that block deployments when retrieval-quality metrics fall below defined thresholds

Researcher & Teaching Assistant

Eötvös Loránd University – Faculty of Informatics

Sep. 2025 – Present

Budapest, Hungary

- Built and optimized DP-SGD federated-learning pipelines for privacy-preserving synthetic data generation ($\epsilon = 1$)
- Developed reproducible data preparation, feature engineering, and evaluation workflows for privacy-utility benchmarking
- Authored and published 13 labs across 8 modules on ML security and privacy 🌐

Software Engineer Intern

ALX Africa

Jun. 2023 – Feb. 2025

Remote

- Designed and implemented backend services and RESTful APIs with database integration in Linux environments
- Analyzed technical specifications and system constraints to architect modular, maintainable software solutions
- Collaborated across Git-based team workflows, adhering to rigorous debugging, unit testing, and documentation standards

PROJECTS

CASPER 🌐

Python, Differential Privacy, RAG, T5, MedCPT, Docker

2026

IDEAL 2026, Main Track

- Built a privacy-preserving RAG framework that clusters sensitive medical records with DP-KMeans and inverts noisy centroids into synthetic prototypes, enabling LLMs to retrieve on private data without accessing raw records
- Evaluated the privacy-utility trade-off, achieving Recall@10 = 0.91 and MIA AUC $\approx$ 0.50 (random-guessing)

Context-Aware Access Control for IoT Devices 🌐

FastAPI, Python, Multi-Agent LLMs, Docker, Wazuh, Prometheus

2026

ITAT'2026, WAS

- Built a hybrid zero-trust IoT authorization framework combining a deterministic validation pre-filter (< 1 ms fast path) with multi-agent LLM reasoning (97 ms slow path) for autonomous context-aware access control
- Achieved 94.1% decision accuracy vs. 82.4% for the RBAC baseline and blocked 100% of 105 red-team adversarial attacks, with fail-secure DENY defaults on LLM rate limits or timeouts

EDUCATION

PhD in Informatics (GPA: 5/5)

Eötvös Loránd University (ELTE)

Sep. 2025 – Expected 2029

Budapest, Hungary

Thesis: Applied synthetic data generation for secure, privacy-preserving AI systems. (Supervised by Pr. Lendák Imre)

M.Sc. in Artificial Intelligence – Ranked 1st in Class (GPA: 3.7/4.0)

University of Khenchela

Sep. 2023 – Jun. 2025

Khenchela, Algeria

Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning. (Supervised by Pr. Dalal Bardou)

B.Sc. in Computer Science – Ranked 1st in Class (GPA: 3.9/4.0)

University of Khenchela

Sep. 2020 – Jun. 2023

Khenchela, Algeria

Thesis: A learning management system for Algerian students 🌐. (Supervised by Dr. Mebarki Djemouai)

CERTIFICATIONS & AWARDS

- **Stipendium Hungaricum Scholarship:** Fully funded PhD studies at ELTE Sep. 2025
- **2nd Place — Starthon'25 Hackathon:** University of El Oued, Algeria Apr. 2025
- **Software Engineering (Back-end Specialization):** ALX Africa 🌟 Feb. 2025
- **AI Career Essentials:** ALX AiCE 🌟 Jul. 2024